

Paper 2 - Set 07 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Functions and transformations
2	D	Integration
3	A	Graphs of functions
4	A	Identifying errors in proofs
5	B	Polynomials, factor and remainder theorems
6	D	Chains of reasoning
7	A	The logic of arguments
8	B	Mathematical proof
9	B	The logic of arguments
10	E	Chains of reasoning
11	H	Binomial expansion
12	D	Chains of reasoning
13	H	Statistics
14	H	Simultaneous equations and inequalities
15	A	The logic of arguments
16	A	Identifying errors in proofs
17	A	Mathematical proof
18	G	Graphs of functions
19	H	Binomial expansion
20	H	Straight lines

Worked solutions

Question 1 FUN-11 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ is transformed by the sequence: reflect in the x -axis, then translate 2 units in the positive y -direction, then reflect in the y -axis. The resulting graph has equation

Answer: D

Reflect in x -axis: $y = -f(x)$. Translate up 2: $y = -f(x) + 2$. Reflect in y -axis (replace x by $-x$): $y = 2 - f(-x)$.

Question 2 INT-20 · Integration · spec MM7.2

Evaluate $\int_{-1}^2 |x| dx$.

Answer: D

Two triangles: $\frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 2 \cdot 2 = \frac{1}{2} + 2 = \frac{5}{2}$.

Question 3 GRA-01 · Graphs of functions · spec MM8.6

How many times does the graph of $y = x^3 - x^2 - 6x$ cross or touch the x -axis?

Answer: A

$x(x^2 - x - 6) = x(x - 3)(x + 2)$: three distinct real roots, so three crossings.

Question 4 ERR-12 · Identifying errors in proofs · spec Err2

A student solves $x^2 = 4x$.

I Divide both sides by x : $x = 4$.

II So the only solution is $x = 4$.

Which of the following is correct?

Answer: A

Factorise instead: $x(x - 4) = 0$ gives $x = 0$ or $x = 4$.

Question 5 X2-08 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x - 2)^2$ is a factor of $p(x)$ '. Which one of the following is true?

Answer: B

If $(x - 2)^2$ divides p then certainly $p(2) = 0$, so P is necessary for Q . But $p(x) = x - 2$ has $p(2) = 0$ without a repeated factor, so P is not sufficient.

Question 6 RSN-14 · Chains of reasoning · spec Prf5

In a class every student studies at least one of Maths and Physics. 70% study Maths and 60% study Physics. A student is picked at random. What is the probability they study both?

Answer: D

$P(M \text{ or } P) = 1 = 0.7 + 0.6 - P(\text{both})$, so $P(\text{both}) = 0.3$.

Question 7 LGC-01 · The logic of arguments · spec Arg1

Consider the statement: ****if n is a multiple of 6, then n is a multiple of 3.**** Which of the following is its contrapositive?

Answer: A

The contrapositive of 'if A then B ' is 'if not B then not A '. It is logically equivalent to the original; B is the converse and C is the inverse.

Question 8 PRF-14 · Mathematical proof · spec Prf1

To prove ****if n^2 is even then n is even**** by contradiction, which assumption should be made?

Answer: B

Assume the hypothesis holds and the conclusion fails, then derive a contradiction: n odd gives n^2 odd, contradicting n^2 even.

Question 9 LGC-18 · The logic of arguments · spec Arg4

A statement reads: ****for every positive integer n , there is a prime p with $n < p < 2n$.** Which of the following is its negation?

Answer: B

Negate the quantifiers in turn: 'for every n , there exists p with ...' becomes 'there exists n such that for every prime p , not ...', i.e. no prime lies strictly between n and $2n$.

Question 10 RSN-09 · Chains of reasoning · spec Prf5

A rectangle has integer side lengths and its area (in cm^2) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting $a \times b$ and $b \times a$ once)?

Answer: E

$$ab = 2a + 2b \Rightarrow (a - 2)(b - 2) = 4: (a - 2, b - 2) = (1, 4) \text{ or } (2, 2), \text{ giving } 3 \times 6 \text{ and } 4 \times 4.$$

Question 11 BIN-12 · Binomial expansion · spec MM2.4

The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?

I $k = 0$

II $k = 3$

III $k = -6$

Answer: H

General term: $\binom{9}{r}(2x)^{9-r}(-x^{-2})^r$ has power $9 - r - 2r = 9 - 3r$, which takes values 9, 6, 3, 0, -3, ..., -18. $k = 0$ needs $r = 3$: yes. $k = 3$ needs $r = 2$: yes. $k = -6$ needs $r = 5$: yes. All three.

Question 12 RSN-03 · Chains of reasoning · spec Prf5

Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?

Answer: D

If four people each shook hands with all four others, everyone shook hands with everyone, so the fifth person also made 4 handshakes, not 1. The others can happen: B is a pentagon of handshakes, C is everyone with everyone, D is a chain, E leaves one person out.

Question 13 X2-28 · Statistics · spec M6.3

A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I The largest value is at least 12.
- II The smallest value is at most 8.
- III The range is at least 4.

Answer: H

The two values above the median are distinct integers greater than 10, so at least 11 and 12 (I); similarly the two below are at most 9 and 8 (II); hence the range is at least $12 - 8 = 4$ (III).

Question 14 SIM-11 · Simultaneous equations and inequalities · spec MM1.5

Given that $|2x - 3| < 5$, which of the following statements is/are true?

- I $-1 < x < 4$
- II $x^2 < 16$
- III $x > -1$

Answer: H

$-5 < 2x - 3 < 5 \Rightarrow -1 < x < 4$, so I is true and III follows from I. For II: every x in $(-1, 4)$ has $x^2 < 16$ (the largest possible $|x|$ is just under 4), so II is also true.

Question 15 LGC-06 · The logic of arguments · spec Arg1

A teacher says: '**You will pass only if you revise.**' Which of the following statements means the same thing?

Answer: A

'A only if B' means ' $A \Rightarrow B$ ': passing implies revising. It does not promise that revising guarantees a pass.

Question 16 ERR-13 · Identifying errors in proofs · spec Err1

Claim: for all positive real x , $x + \frac{1}{x} \geq 2$.

- I Suppose $x + \frac{1}{x} \geq 2$.
- II Multiply by $x > 0$: $x^2 + 1 \geq 2x$.
- III So $x^2 - 2x + 1 \geq 0$, i.e. $(x - 1)^2 \geq 0$, which is true.
- IV Hence the claim is proved.

Which of the following best describes the argument?

Answer: A

Starting from the conclusion and reaching a truth does not prove the conclusion unless every step is reversible and the argument is presented in the forward direction. Here the steps are reversible, so the repair is to write it backwards.

Question 17 PRF-09 · Mathematical proof · spec Prf1

Here is a proof that there are infinitely many primes, with the steps out of order.

- I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .
- II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \cdots p_n + 1$.
- III So N has a prime factor that is not in the list, which is a contradiction.
- IV But $N > 1$, so N has at least one prime factor.

Which is the correct order?

Answer: A

Define N (II), show no listed prime divides it (I), note that N nevertheless has some prime factor (IV), and conclude that this factor is a new prime (III). The conclusion III cannot come before both I and IV have been established.

Question 18 X2-21 · Graphs of functions · spec MM8.6

$p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I $p(x) = 0$ has at least one real root.
- II $p(x) = 0$ has at most four real roots.
- III The graph of $y = p(x)$ has at least one turning point.

Answer: G

I fails: $x^4 + 1$ has no real roots. II: a degree-4 polynomial has at most 4 roots. III: an even-degree polynomial tends to the same infinity at both ends, so it must turn at least once.

Question 19 X2-13 · Binomial expansion · spec MM2.4

n is a positive integer. Which of the following statements is/are true?

- I The coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$.
- II The sum of the coefficients in the expansion of $(1-x)^n$ is 0.
- III $(1+x)^n \geq 1+nx$ for all $x \geq 0$.

Answer: H

I is the general term with $r = n$. II: put $x = 1$. III: expanding $(1+x)^n$ gives $1+nx$ plus non-negative terms when $x \geq 0$.

Question 20 LIN-10 · Straight lines · spec MM3.1

The line through $(a, 0)$ and $(0, b)$, with $a, b > 0$, passes through $(2, 3)$. Which of the following statements is/are true?

- I $\frac{2}{a} + \frac{3}{b} = 1$
- II $a > 2$ and $b > 3$
- III The smallest possible value of ab is 24

Answer: H

The line is $\frac{x}{a} + \frac{y}{b} = 1$, so I holds. Since $\frac{2}{a} < 1$ and $\frac{3}{b} < 1$, II holds. $b = \frac{3a}{a-2}$, so $ab = \frac{3a^2}{a-2}$; setting $a = 2+t$ gives $\frac{3(2+t)^2}{t} = 3(\frac{4}{t} + 4 + t) \geq 3(4+4) = 24$ by AM-GM (equality at $t = 2$, $a = 4$, $b = 6$). III holds.